

Q1. Enable versioning on the personal column family — keep up to 5 versions.

```
File Edit View Search Terminal Help
hbase(main):020:0> alter 'student', {NAME => 'personal', VERSIONS => 5}
Updating all regions with the new schema...
1/1 regions updated.
Done.
Took 1.8739 seconds
hbase(main):021:0> describe 'student'
Table student is ENABLED
student
COLUMN FAMILIES DESCRIPTION
{NAME => 'academic', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '3',
KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => '
NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}

{NAME => 'contact', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1',
KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => 'N
ONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}

{NAME => 'personal', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '5',
KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => '
NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}
```

```
File Edit View Search Terminal Help
Table student is ENABLED
student
COLUMN FAMILIES DESCRIPTION
2. {NAME => 'academic', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '3',
KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => '
NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}

{NAME => 'contact', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1',
KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => 'N
ONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}

{NAME => 'personal', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '5',
KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => '
NONE', TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}

3 row(s)

QUOTAS
0 row(s)
11 Took 0.0600 seconds_
```

Q2. Update personal:city for student 2 (sara) three times: first to cairo, then giza, then back to alex. Then retrieve all versions of that cell.

```
QUOTAS
0 row(s)
Took 0.0600 seconds
hbase(main):022:0> put 'student', '2', 'personal:city', 'cairo'
Took 0.0075 seconds
hbase(main):023:0> put 'student', '2', 'personal:city', 'giza'
Took 0.0106 seconds
hbase(main):024:0> put 'student', '2', 'personal:city', 'alex'
Took 0.0050 seconds
hbase(main):025:0> get 'student', '2', {COLUMN => 'personal:city', VERSIONS => 5
}
COLUMN          CELL
personal:city    timestamp=2026-05-24T10:31:34.423, value=alex
personal:city    timestamp=2026-05-24T10:31:27.379, value=giza
personal:city    timestamp=2026-05-24T10:31:19.772, value=cairo
personal:city    timestamp=2026-05-23T13:53:57.333, value=alex
1 row(s)
Took 0.0148 seconds
hbase(main):026:0>
```

Q3. Use SingleColumnValueFilter to find all students in year 4.

```
bigdata@localhost:~
File Edit View Search Terminal Help
hbase(main):001:0> scan 'student', {
hbase(main):002:1* FILTER => "SingleColumnValueFilter('academic','year',=,'binary:4')"}
are: ROW          COLUMN+CELL
2      column=academic:faculty, timestamp=2026-05-23T13:56:25.321, value=medicine
2      column=academic:grade, timestamp=2026-05-23T13:57:22.832, value=B
2      column=academic:year, timestamp=2026-05-23T13:58:17.493, value=4
2      column=contact:email, timestamp=2026-05-24T10:26:28.212, value=sara@uni.edu
2      column=personal:age, timestamp=2026-05-23T13:55:16.665, value=22
2      column=personal:city, timestamp=2026-05-24T10:31:34.423, value=alex
2      column=personal:name, timestamp=2026-05-23T13:52:34.475, value=sara
4      column=academic:faculty, timestamp=2026-05-23T13:56:43.637, value=arts
4      column=academic:grade, timestamp=2026-05-23T13:57:44.456, value=C
count 4      column=academic:year, timestamp=2026-05-23T13:58:40.610, value=4
```

```
File Edit View Search Terminal Help
2      alue=4
      column=contact:email, timestamp=2026-05-24T10:26:28.212, v
      alue=sara@uni.edu
are: 2      column=personal:age, timestamp=2026-05-23T13:55:16.665, va
      lue=22
2      column=personal:city, timestamp=2026-05-24T10:31:34.423, v
      alue=alex
2      column=personal:name, timestamp=2026-05-23T13:52:34.475, v
      alue=sara
4      column=academic:faculty, timestamp=2026-05-23T13:56:43.637
      , value=arts
4      column=academic:grade, timestamp=2026-05-23T13:57:44.456,
      value=C
4      column=academic:year, timestamp=2026-05-23T13:58:40.610, v
      alue=4
4      column=personal:age, timestamp=2026-05-23T13:55:46.632, va
      lue=23
gui 4      column=personal:city, timestamp=2026-05-23T14:06:10.642, v
      alue=mansoura
4      column=personal:name, timestamp=2026-05-23T13:52:59.348, v
      alue=nada
2 row(s)
bunt Took 0.7936 seconds
```

Q4. Use SingleColumnValueFilter to find all students who live in cairo.

```
hbase(main):003:0> scan 'student', {
hbase(main):004:1* FILTER => "SingleColumnValueFilter('personal','city',=,'binar
y:cairo')"}
ROW      COLUMN+CELL
1      column=academic:faculty, timestamp=2026-05-23T13:56:14.238
      , value=engineering
1      column=academic:grade, timestamp=2026-05-24T10:23:40.235,
      value=A+
1      column=academic:year, timestamp=2026-05-23T13:58:06.547, v
      alue=3
1      column=contact:email, timestamp=2026-05-24T10:26:17.936, v
      alue=ali@uni.edu
1      column=personal:age, timestamp=2026-05-23T13:55:02.007, va
      lue=21
1      column=personal:city, timestamp=2026-05-23T13:53:47.328, v
      alue=cairo
1      column=personal:name, timestamp=2026-05-23T13:50:10.879, v
      alue=ali
1 row(s)
Took 0.0415 seconds
hbase(main):005:0>
```

Q5. Add contact:phone for students 1, 3, and 4 with any phone numbers you choose.

```
hbase(main):005:0> put 'student', '1', 'contact:phone', '01012345678'
Took 0.0562 seconds
hbase(main):006:0> put 'student', '3', 'contact:phone', '01198765432'
Took 0.0076 seconds
hbase(main):007:0> put 'student', '4', 'contact:phone', '01255554444'
Took 0.0063 seconds
hbase(main):008:0>
```

Q6. Scan the table and return only the contact column family for all students.

```
hbase(main):008:0> scan 'student', {COLUMNS => 'contact'}
ROW          COLUMN+CELL
1            column=contact:email, timestamp=2026-05-24T10:26:17.936, v
             alue=ali@uni.edu
1            column=contact:phone, timestamp=2026-05-24T10:36:49.533, v
             alue=01012345678
2            column=contact:email, timestamp=2026-05-24T10:26:28.212, v
             alue=sara@uni.edu
3            column=contact:phone, timestamp=2026-05-24T10:36:57.656, v
             alue=01198765432
4            column=contact:phone, timestamp=2026-05-24T10:37:04.432, v
             alue=0125554444
4 row(s)
Took 0.0140 seconds
```

Q7. Add a new column family called enrollment to the student table, then verify it appears in the schema.

```
File Edit View Search Terminal Help
Took 0.0140 seconds
hbase(main):009:0> alter 'student', NAME => 'enrollment'
Updating all regions with the new schema...
1/1 regions updated.
Done.
Took 3.3107 seconds
hbase(main):010:0> describe 'student'
Table student is ENABLED
student
COLUMN FAMILIES DESCRIPTION
{NAME => 'academic', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '3',
KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => 'NONE',
TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}

{NAME => 'contact', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1',
KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => 'NONE',
TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}

{NAME => 'enrollment', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1',
KEEP_DELETED_CELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => 'NONE',
TTL => 'FOREVER', MIN_VERSIONS => '0', BLOCKCACHE => 'true', BLOCKSIZE =>
'65536', REPLICATION_SCOPE => '0'}
```

Q8. CHALLENGE : Write a scan that returns only students who are in the engineering faculty AND in year 3.

```
Took 0.0357 seconds
hbase(main):013:0> scan 'student', {FILTER => "SingleColumnValueFilter('academic', 'faculty', '=', 'binary:engineering') AND SingleColumnValueFilter('academic', 'year', '=', 'binary:3')"}
ROW                                COLUMN+CELL
1                                  column=academic:faculty, timestamp=2026-05-23T13:56:14.238, value=engineering
1                                  column=academic:grade, timestamp=2026-05-24T10:23:40.235, value=A+
1                                  column=academic:year, timestamp=2026-05-23T13:58:06.547, value=3
1                                  column=contact:email, timestamp=2026-05-24T10:26:17.936, value=ali@uni.edu
1                                  column=contact:phone, timestamp=2026-05-24T10:36:49.533, value=01012345678
1                                  column=personal:age, timestamp=2026-05-23T13:55:02.007, value=21
1                                  column=personal:city, timestamp=2026-05-23T13:53:47.328, value=cairo
1                                  column=personal:name, timestamp=2026-05-23T13:50:10.879, value=ali
1 row(s)
Took 0.0798 seconds
hbase(main):014:0>
```